Koa Shen

Mechanical Engineering | Manufacturing, DFM & Precision Fabrication | Summer 2027

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EDUCATION

University of California, Santa Barbara Expected June 2028

B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94

- Dean's Honors every quarter. Relevant coursework: Mechanics of Materials, Dynamics, Circuits, MATLAB; Dynamical Systems in progress.

MANUFACTURING & ENGINEERING EXPERIENCE

Mechanical Engineering Intern Summer 2026 - Present

SLAC National Accelerator Laboratory, LCLS | Menlo Park, CA; extended part-time remote

- Produced Solid Edge designs and PDM-controlled documentation with GD&T for manufacture and inspection of beamline motion assemblies.
- Redesigned optics packaging, kinematic mounts, alignment features, and detector strain relief; checked load cases to address crash and drop risks for $100k+ detectors.
- Built a Python/Drake digital twin for a 30-DOF, 305-body assembly; millisecond collision checks identified at least five pinch points and guided design changes.
- Created 15 reusable motion-stage models; the framework is being adopted across three LCLS assemblies.

Fabrication Team Lead Fall 2024 - Present

Gaucho Racing, Formula SAE Electric | UC Santa Barbara

- CNC-machined 33 precision suspension components from 4130 steel and 7075-T6 aluminum on Haas mills and lathes; held clevis and rod-end bores within a 0.0004 in tolerance window and bearing interfaces within 0.0005 in, verified with go/no-go tests.
- Worked with suspension designers on GD&T and manufacturability; proposed a standardized spindle interface to reuse CAM programs and simplify setup and programming.
- Led 15 teammates through driver safety-system design, manufacturing, and installation; GR26 passed FSAE technical inspection fully rules-compliant.
- Own chassis and ergonomics packaging, coordinating welded tube structures, assembly interfaces, and manufacturing across vehicle subteams.

Maintenance Technician Summer 2025

Exploratorium | San Francisco, CA

- Serviced 100+ public exhibits and completed three major overhauls; redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement-part costs.
- Reverse-engineered exhibit parts and developed CAD, CAM, and fixturing for replacement components; documented remanufacturing instructions for future technicians.

SELECTED MANUFACTURING PROJECTS

Formula SAE Steering-Wheel Development 2024 - 2026

- Helped reduce assembly mass from 5 lb to 1.7 lb through iterative design and FEA-guided optimization; coordinated manufacturing and integration of buttons, potentiometers, a display, and quick-release hardware.

Compact Cycloidal Actuator Summer 2026 - Present

- Designed and fabricated a 15:1 reducer within a NEMA 17 motor's 42 mm square footprint; selected PETG/ABS print orientations for load capacity and dimensional accuracy.

TECHNICAL SKILLS

Manufacturing: Haas CNC mills and lathes, Mastercam, CAM, manual machining, welding, fixturing, 3D printing, assembly

Design & Quality: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, FEA, DFM, go/no-go inspection, design documentation

Analysis & Software: Python, C/C++, MATLAB, Drake, kinematics, statics/dynamics, Git